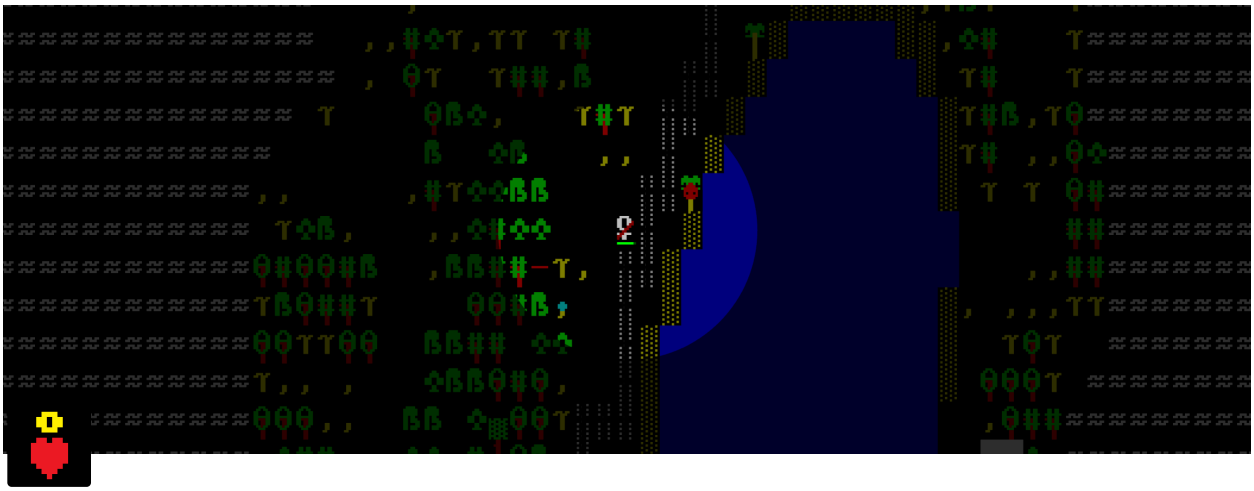


ALIVE - Game Design

ASCII-based 2D adventure game

- Play it here: https://ab-und-zu.github.io/alive/
- Source code: https://github.com/AB-UND-ZU/alive





(A) Game Design

Main pillars 🚧

- Something to get instantly hooked, like initial quest or story line
- Should feel like living inside a OS terminal
- Mutable, ephemeral, generated 2D ASCII worlds

- Simplified controls: only directional keys and two actions (primary, secondary)
- Fog of war discovery, tiling world boundaries
- Creative spells and fighting dynamics, limited possibilities due to strong 2D grid
- Gathering, crafting, enchanting to fight stronger mobs and enter new realms
- Go on adventures outside, then return to city to recover, buy and craft
- Three main elements: fire for burning, water for freezing, earth for healing
- Three tribes that the player can join, unlocks items, makes others hostile
- Generally, any goal should be achievable by either farming a lot, or high skilled PvE
- Choice to cooperate or fight NPCs, e.g. killing and stealing instead of trading
- Parity between all platforms, portable to basically anything

Story 💡

TBD

Game modes

▼ Single player adventure 🚧

- Start in “hills” world which has elements from each tribe world
- Each world has a few main quests and bosses which will drop keys to next worlds
- Player is able to join a tribe or stay neutral, leaving 4 possible paths to take
- Each path at some point has an infinite waves arena which could serve as a high score

▼ Multiplayer 💡

- Players start in a neutral world which realms they can enter (RotMG)
- Within the realms they can acquire items and join a tribe
- Optional PvP free-for-all realms with better item drops
- One global tribes realm where cities fight against each other and can be captured, defeating the chief

Inspiration

▼ Realm of the Mad God (RotMG)

> Inspiring aspects: **2D top down, classes and spells, ephemeral realms, world bosses**

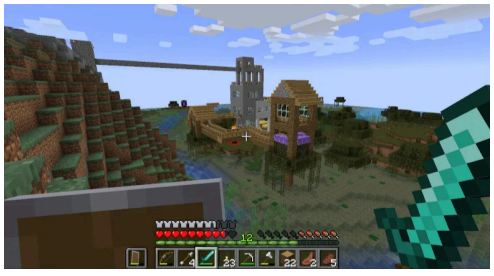


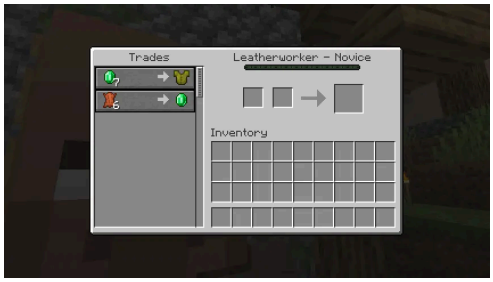


- Link: https://www.realmofthemadgod.com/
- MMORPG, bullet-hell, perma death, roguelike
- 2D top down view, rotatable, blocky pixel art, terrain in grid but walking freely
- Different classes with one weapon and one spell but some variations
- Leveling 1 to 20 and 8 stats with max caps, tiered items from mobs and dungeons
- Ephemeral realms are running for around 1 hour and have world events and global bosses
- No farming or crafting, all PvE and grinding
- Massively fun and addictive (until you die, but it unlocks new classes)
- Freemium, skins but also in recent years added items and pets for money

▼ Minecraft (MC)

Inspiring aspects: **terrain generation, biomes, farming, trading system, enchanting**





- Sandbox survival, world builder, co-op play, procedurally generated
- 3D blocky pixel art, infinite terrain in grid but free movement
- Overworld, Nether, End, each with different biomes

- No classes, player chooses tools/weapons/potions for combat or farming
- No traditional leveling, progression through gear, crafting tiers, and rare items
- Enchanting system, trading system with villager NPCs, tamable pets
- Heavy focus on farming, crafting, building
- Endlessly replayable, large creative freedom, but fighting sucks

▼ Zelda: Breath of the Wild (BotW)

> Inspiring aspects: **quest driven, exploration, sense of progression, easy to start**





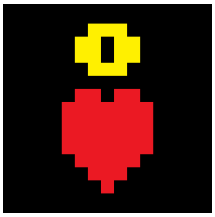
- Open-world action-adventure, many RPG elements, story driven
- 3D semi-realistic cartoon, huge map with physics-based systems
- No classes, free-form weapon and armor usage, weapons break over time, wide variety of combat styles
- Progression via heart containers, stamina upgrades, better gear, and shrine puzzles
- No crafting, but gathering, cooking, and weapon collecting are key
- Very immersive, encourages exploration over strict questing

▼ (B) Branding

Name 🚧

Current working title is “Alive”, since this was the name of the initial version of the game as the first program I ever wrote (as in “Hello world”)

Logo 🚧

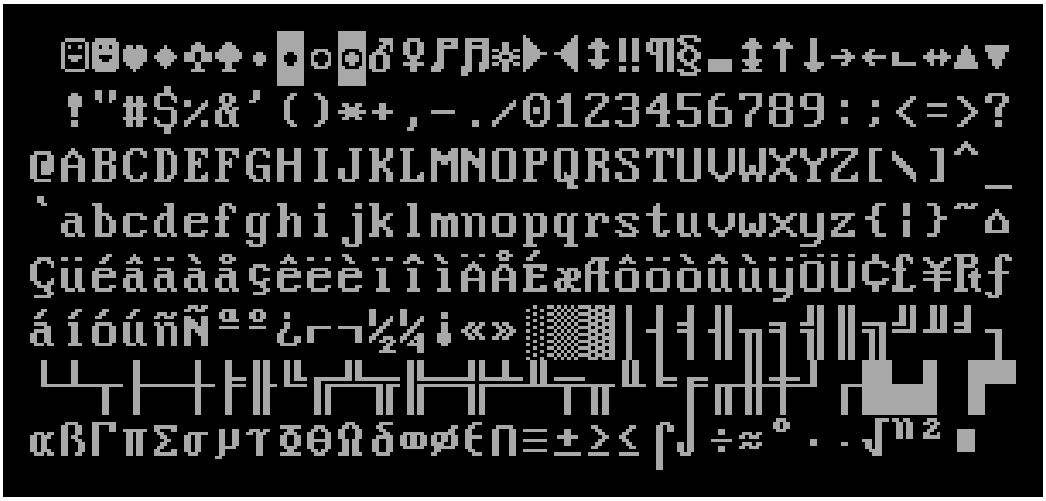


A simple arrangement of the ASCII heart and degree character, symbolizing a halo

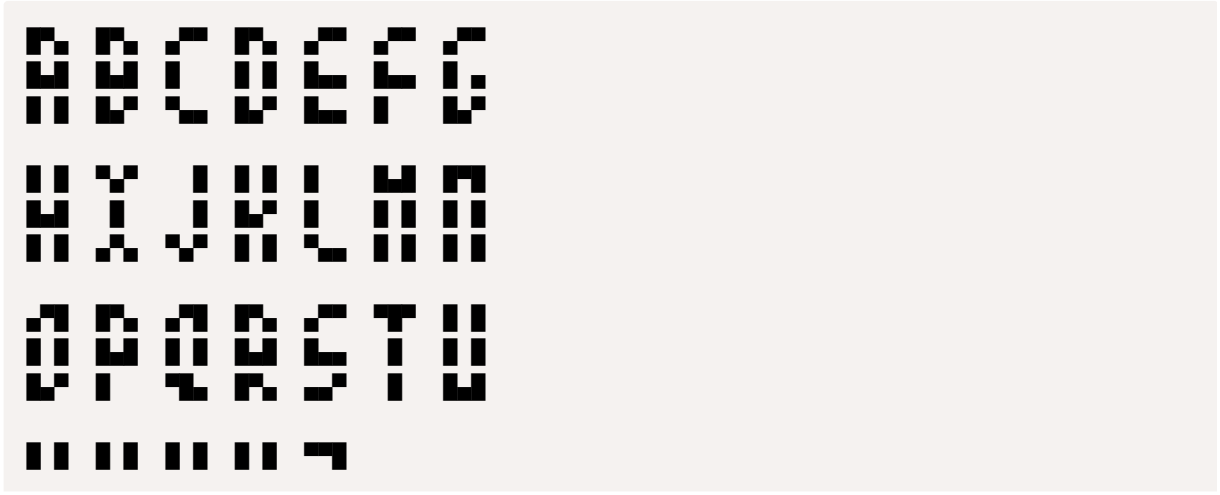
Font

Only the 256 characters in code page 437 from the "DOS Font" are used (available here: https://laemeur.sdf.org/fonts/), along with original 16 terminal colors. To create all sprites, characters are stacked over each other (with the only exception of health bars and stack counters, they are underscores `_` transformed to the appropriate length and dots, respectively)

“DOS Font”



Large font 3x3 🚧



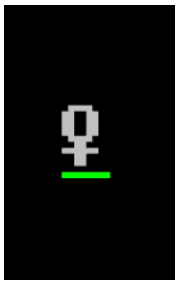


(C) Characters & Entities

▼ Classes 🚧

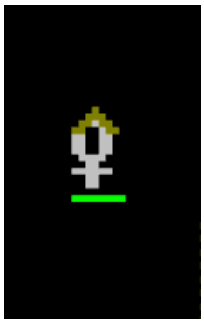
Each class gives one specific base attribute, but can still use all equipment and spells

Scout



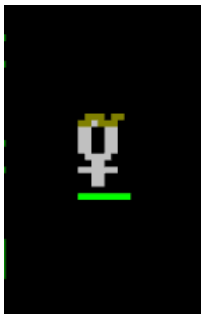
- +1 melee dmg
- Medium HP
- Medium mana
- Neutral tribe

Knight



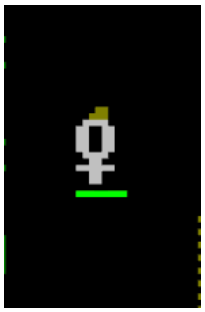
- +1 defense
- High HP
- Low mana
- Can join tribe

Mage



- +1 magic dmg
- Low HP
- High mana
- Can join tribe

Hunter



- +1 speed
- Low HP
- Medium mana
- Can join tribe

▼ Tribes 💡

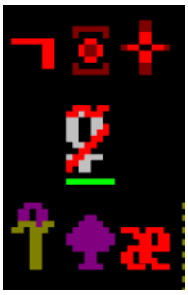
Neutral tribe is friendly with all tribes. Fire, water and earth are hostile to each other

Tribe of character and NPC is visible through hair color. Neutral NPCs have white hair, neutral players are blonde. NPCs in a tribe town have all same hair color.

Fire

Water

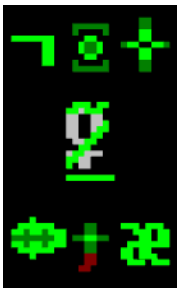
Earth



- Burn units and plants
- Damage over time
- Thaw ice
- Fire essence made from berries
- Own world with desert and volcanoes



- Freeze units and water
- Stuns units
- Extinguish fire
- Water essence made from flowers
- Own world with ocean and glaciers

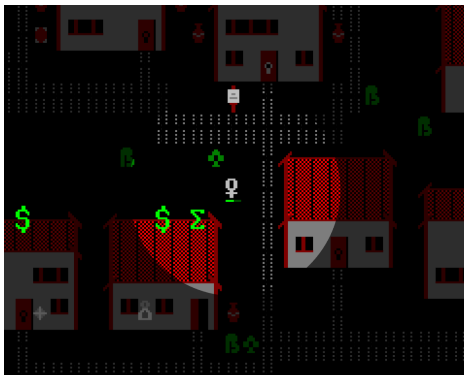


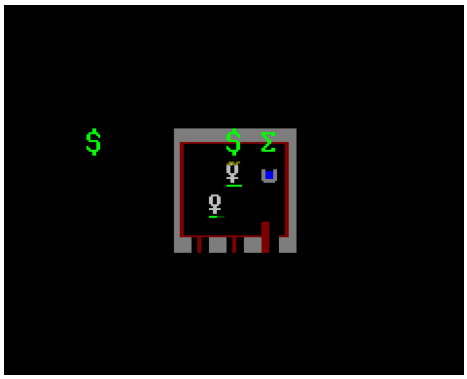
- Heal oneself and allied units
- Earth essence made from seeds
- Own world with jungle and caves

▼ NPCs

NPCs offer quests, shops or crafting

NPCs wear a combination of equipment and hair



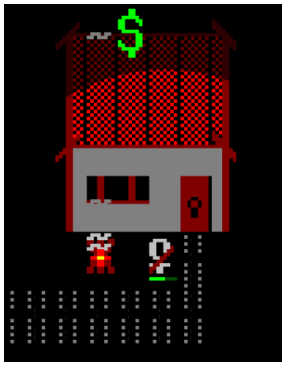


Guide 🚧

Nomad 🚧

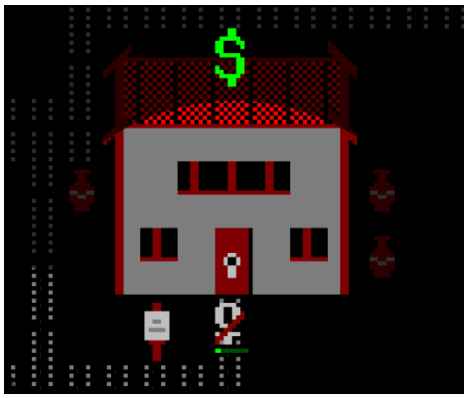


- Gives intro quest
- Maybe also final boss?



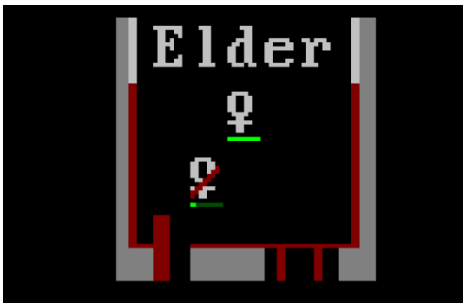
- House outside of city
- No tribe
- Trades key for chief
- Can be killed and key stolen

Chief 🚧



- Each tribe has a chief in center of town
- Spawn point in front of its house
- Locked iron door, needs to find Nomad
- Access to dungeon or next world
- Sign with quest in front of house

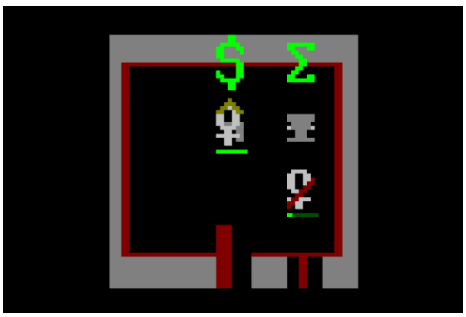
Elder 🚧

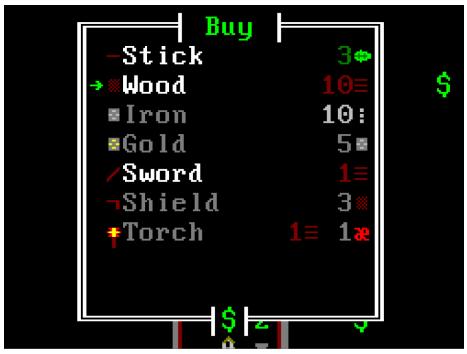


- Gives various gather and kill quests
- Quest for world boss

Smith

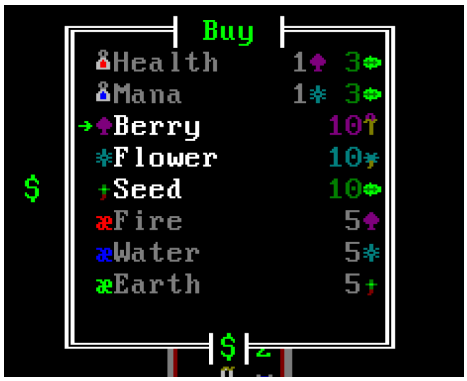
Anvil 🚧





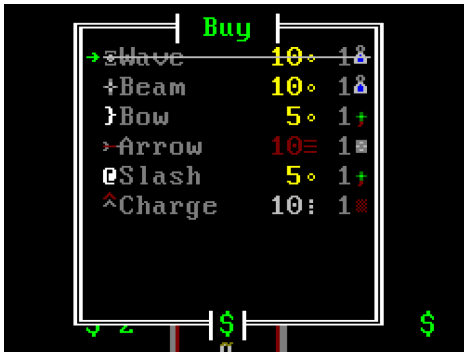
- Shop for resources and equipment
- Next to anvil for crafting

Druid



- Shop for potions and essences
- Next to kettle for enchanting

Mage

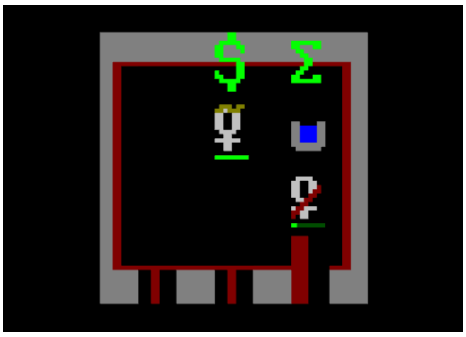


- Shop for spells and abilities

Scout

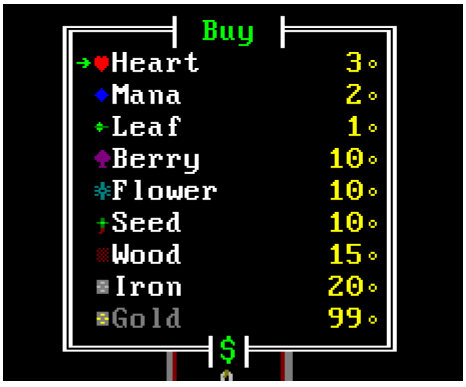
- Enchanting equipment with elements
- Crafting higher tiered equipment with resources such as iron or gold

Kettle 🚧



- For crafting elemental spells and items

Trader



- Buy various materials for gold

Fisher 💡

- Near lake outside city
- Shop for fishing rod and worms



- Ability to sell item drops for gold
- Gold can be used to buy other materials

Miner 💡

- Near mountain outside city
- Shop for pickaxe and ores

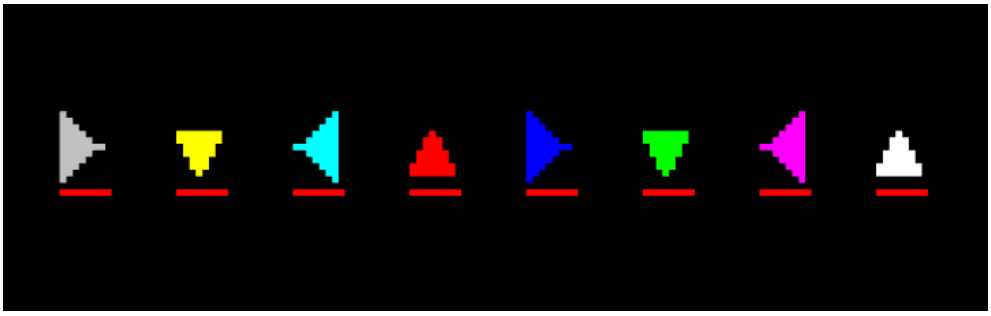
Ranger 💡

- Near forest outside city
- Shop for axe and materials

▼ Mobs

Mobs are hostile to all units (except themselves), have a specific behaviour and drops

Levels 🚧

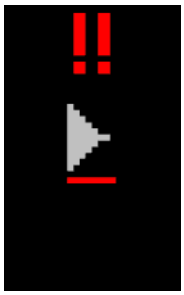


- **Iron, Gold, Diamond, Ruby**: stronger with each level
- **Fire, Water, Earth**: elemental on-hit effects
- **White**: placeholder for epic tier with animations

Prism

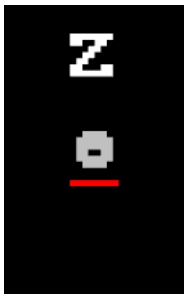
Eye

Orb



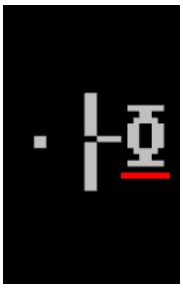
- High HP
- Low dmg
- Walks in straight line
- Turns randomly when hitting walls
- Shows rage when player in walking line
- Always moves

Fairy



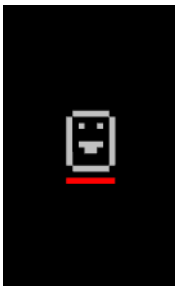
- Low HP
- High dmg
- Sleeps when hidden
- Alert when player is close
- Rage when too close
- Simple directional walking towards player

Bandit 💡



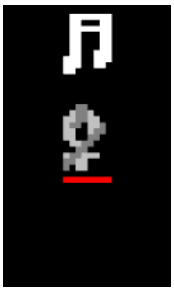
- Medium HP
- Ranged dmg
- Half speed movement
- Casts beam spell
- Shoots into walking line of player
- Shows rage before casting
- Flees from player, otherwise standing

Shark 💡



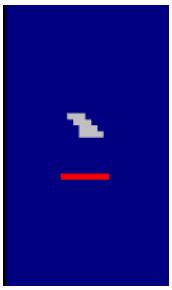
- High HP
- No dmg
- Flees far from player
- Side steps randomly when close
- Drops essences

Cube 💡



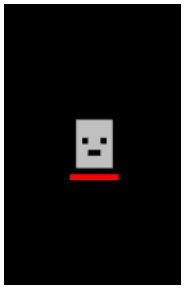
- Medium HP
- High dmg
- Whistles, sits around campfire
- Protects a chest
- Follows player for a bit

Blade 💡

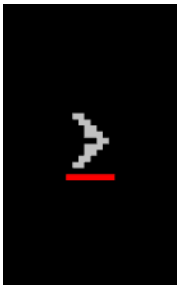


- Medium HP
- High dmg
- Circles around lake

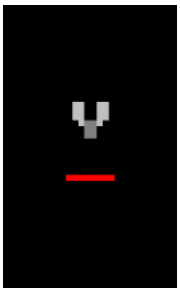
Fly 💡



- High HP
- Ranged dmg
- Casts wave spell
- Shows rage before casting
- Jumps as movement
- Can jump over terrain
- Gets closer to player

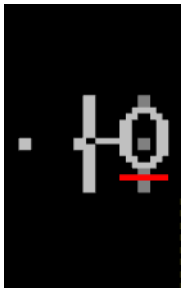


- Medium HP
- Spell damage
- Charges when player in linear direction
- Invincible while charging
- Spinning and stunned when hitting terrain



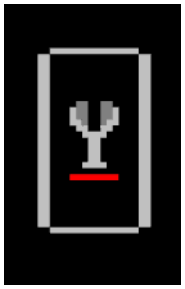
- Low HP
- Medium dmg
- Can fly through terrain
- Sits on plants
- Random movement towards player
- Spawns in groups

Beam tower 💡



- High HP
- High ranged dmg
- Rage before casting
- Casts beam spell
- No movement
- Usually in dungeons

Wave tower 💡



- High HP
- High ranged dmg
- Rage before casting
- Casts wave spell
- No movement
- Usually in dungeons

▼ Bosses 💡

- Frozen statues that awake if hit by player
- World bosses wandering around
- Special mini biome for boss in world
- Dungeon bosses with phases
- Sprites could span multiple cells

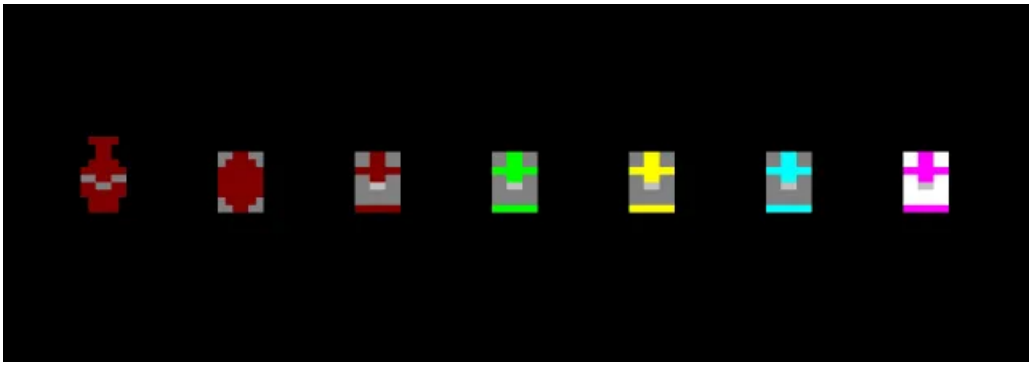
- TBD
- House boss
- Chest boss
  - Water and campfire in center
  - Elemental wave towers in corners affect water and fire
  - Spell pattern from top dodge
  -
- Tree boss

▼ Units

> Neutral units that can be destroyed and drop items

Grey health bars are only shown once they are hit

Levels 🚧

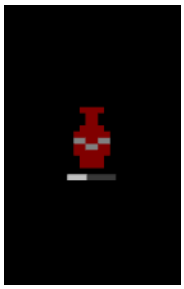


- Container units have increasingly better item drops but also more HP with higher levels

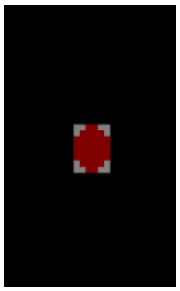
Pot

Box

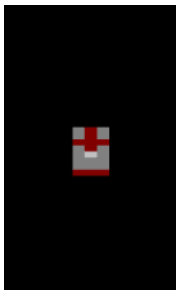
Chest



- Medium HP
- Drops 1x HP/MP or 3x coin/berry/flower/leaf/ore/stic
- Next to houses

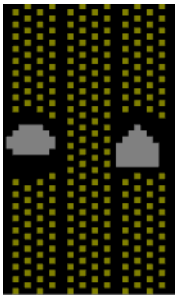


- Low HP
- Can be pushed around
- Can't be hit with sword
- Next to houses
- Drops 1x XP or stack of berry/flower/seed, or 3x apple/shroom/coconut/banar



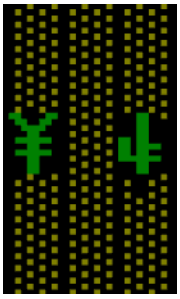
- High HP
- Increasingly better items with levels
- Usually protected by an NPC or mob

Rocks



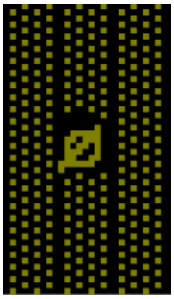
- High HP
- Appears in desert
- Drops ore, or gem/crystal

Cacti



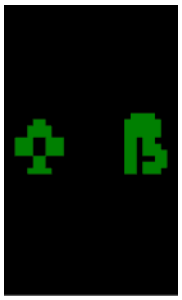
- Medium HP
- Spike damage on hit
- Appears in desert
- Drops seed

Tumbleweed



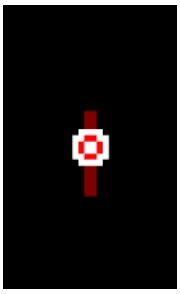
- Low HP
- Appears in desert
- Moves to the right and spins
- Teleports back to left once at end of desert
- Drops stick

Hedges



- Medium HP

Target 💡



- Infinite HP

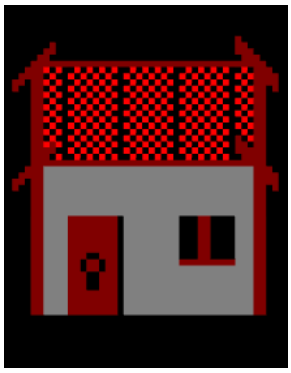
- Appears in forests and general areas
- Can sometimes allow passing closed terrain
- Drops leaf
- No drop if burned

- Heals itself
- Can be used to test DPS, or as target for quests
- No drops

▼ Structures

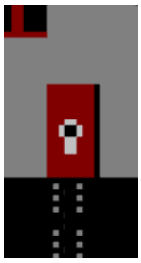
Structures can be interacted with but can’t usually be destroyed

Houses



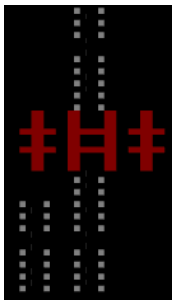
- Can be entered
- Render differently once inside
- Usually contains NPC
- Enclosed by walls and door
- Composed of multiple segments

Doors



- Consumes key to unlock
- Once opened, will remain unlocked
- Keys in different materials
- Mobs can’t pass, even when open

Fence gate 🚧

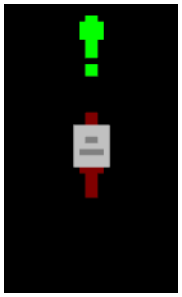


- Can be opened
- No key required
- Can be burned
- Mobs can’t pass, even when open

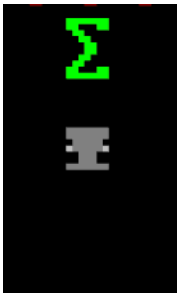
Sign

Anvil 🚧

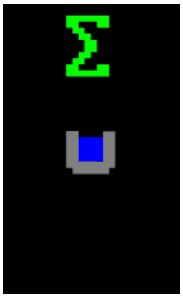
Kettle 🚧



- May offer tips
- Usually gives a quest

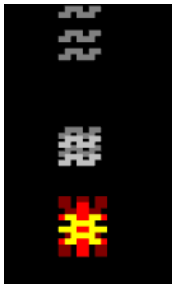


- Can be used to craft equipment
- Uses wood and ore materials



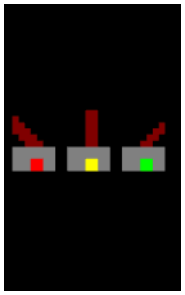
- Can be used to craft elemental spells
- Can also enchant equipment

Fire



- Burns all units on it
- Eternal fire
- Always visible fire and smoke
- Can be extinguished
- Can be re-ignited

Lever 💡



- Can be switched by player
- Can be used for quests or settings

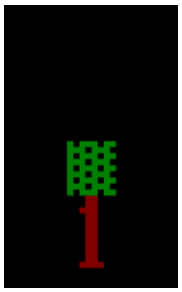
▼ Plants

Plants may carry fruit, and can be burned with fire spells, otherwise act as terrain

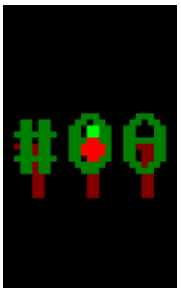
Large tree

Trees

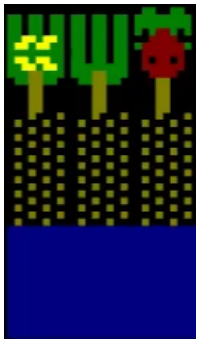
Palms



- Two cells tall
- Drops wood when chopped
- Burns whole tree if one part catches fire
- Longer burn duration

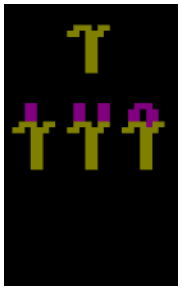


- Drops stick when chopped
- Can be burned
- May carry an apple



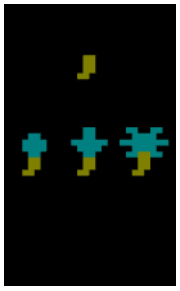
- Near beaches
- Drops stick when chopped
- Can be burned
- May carry a banana or coconut

Bush



- Walkable terrain
- May carry 1-3 berries
- Burns very quick

Grass



- Walkable terrain
- May carry 1-3 flowers
- Burns very quick

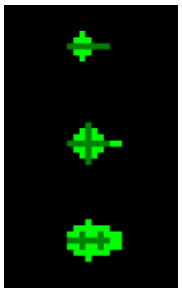
▼ Drops

Drops are naturally occurring items laying around that can be picked up

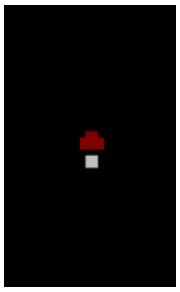
Leaves

Shroom

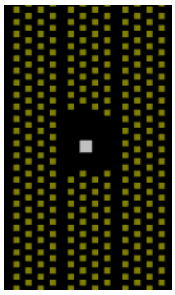
Stone



- Appears in 1-3
- Close to grass and bushes

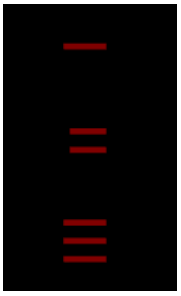


- Appears single only
- Close to trees and hedges



- Appears single only
- In desert biome

Sticks

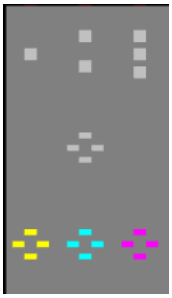


- Appears in 1-3
- Close to trees and hedges

▼ Terrain

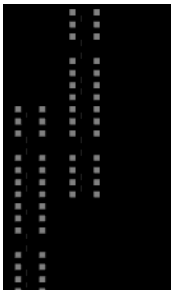
Terrain can rarely be directly affected by players, but may still have an effect on them

Mountain



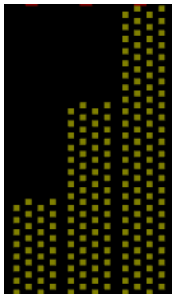
- Casts shadow

Path



- Leads path between waypoints

Sand

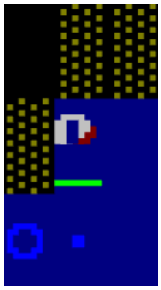


- May carry 1-3 ore
- Can have iron, gold, diamond or ruby mine
- Can be mined with pickaxe

- Speeds up player significantly
- No effect for mobs or NPCs

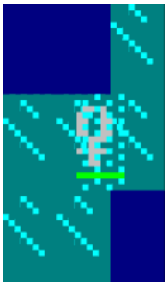
- Near water, or part of desert biome
- Slows down player
- No effect for mobs or NPCs

Water



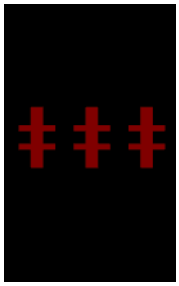
- Makes units and items swim inside
- Only walkable if within one cell reach of land
- Slows down player significantly
- No effect for mobs or NPCs
- Extinguishes if burning
- Can be frozen
- Creates bubbles when walking through
- Can be passed with boat

Ice



- Slippery surface, causes units to slide into direction they entered the ice
- Stunned while sliding
- Can be thawed with fire
- Stops sliding once hitting wall or water

Fence



- Can't be passed
- Can be burned
- Protecting a town

Lava 💡

- Slows down movement extremely
- Causes unit to burn
- Can be turned to stone with water spell

Quicksand 💡

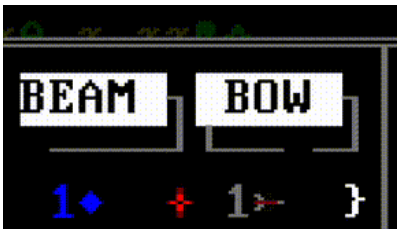
- Slows down movement extremely
- Can sink player and cause suffocation damage
- Can be turned to sand with fire spell

(D) Dynamics

Systems (ECS)

A system in ECS operates on all entities with given components. In this context, they handle low to medium frequency logic.

▼ Primary and secondary actions



Available primary and secondary actions are determined by player position and adjacent entities. Possible interactions with NPCs and structures take precedence over actionable items. Only one or less primary and secondary action each is possible at any given point in time.

The priority list is as follows:

Condition	Primary	Secondary
1. Player is dead	[SPAWN]	-
2. Player is adjacent to quest	[QUEST]	-
3. Player is adjacent to door	[OPEN]	-
4. Player is adjacent to shop or crafting	[SHOP] or [CRAFT]	-
5. Player has opened shop or craft	[BUY] or [SELL] or [CRAFT]	[CLOSE]
6. Player has actionable items	[<PRIMARY>] like [WAVE]	[<SECONDARY>] like [BOW]
7. Otherwise	-	-

Primary actions can be triggered by tapping or clicking the left button, or by pressing **SPACE** or **ENTER**. They usually change state or have a cooldown, such as waiting for a spell to finish. As for items, they are more powerful and consume mana.

Secondary actions can be triggered by tapping or clicking the right button, or by pressing **SHIFT** or **TAB**. They usually can be triggered multiple times in a row. As for items, they are less powerful but have an iten based system which allows for regeneration, such as collecting previously shot arrows, or receiving new charges upon killing enemies with melee attacks.

Only one action can be triggered at a time. That includes movements, meaning that if a primary or secondary action is pressed, no movement will be executed in this tick.

▼ Behaviour of NPCs, mobs and units

All NPCs and mobs interact in the same interval, the world tick. Therefore they are generally not affected by terrain which increases or decreases speed.

When following quests, an NPC can queue multiple behaviors, for example walk to the house, open the door, pick up an item, close the door, walk back, show a dialog. Outside interferences may interrupt the behaviours, possibly causing them to be dropped or replaced, for example if the player dies or an item is stolen.

Each mob follows one pattern which is influenced by different conditions such as player proximity or movement:

Mob/Unit	Behaviour
Prism	- If a player is within a few walkable blocks in the current direction, show rage symbol - Attempt to walk or attack in currently facing direction - If a wall is encountered, turn into the closest free direction - If all walls are blocked (prism is trapped), spin randomly - Always keep moving
Eye	- Show sleeping symbol and stand still - If player comes close, open eyes and show alert symbol - If player comes even closer, show rage symbol and start walking - Walk in directions closest to angle towards player (that means it might just get stuck if a wall is right between player and Eye) - If player runs too far away, or Eye is in shadow, fall back asleep
Orb	- Stand still and wait - If a player is walking towards a linear direction, cast beam spell early to hit in its path - If a player is directly aligned in a linear direction, cast beam spell - Before casting beam spell, shortly show rage symbol - If player gets too close, flee away from player in opposite angle - No attacks while player is too close - Movement only every two ticks
Fairy	- Stand still and wait - If player is in medium range, flee away from player in opposite angle - Try to flee to the sides if fleeing direction is facing a wall - The closer the player gets, the more often randomly sidestep to avoid being attacked - If Fairy is in trajectory of an arrow shot by player, sidestep to avoid being hit
Tumbleweed	- Move to the right - Once reaching the end of the desert, disappear - After a randomized amount of time, teleport to the leftmost cell of the cell at the same height - Restore all health, reappear and continue moving

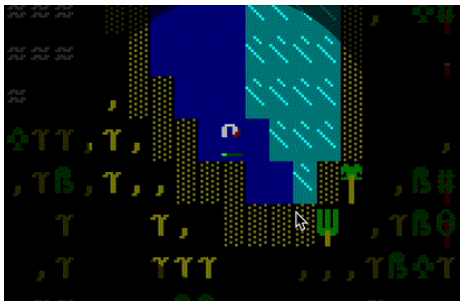
▼ Ballistics of arrows



Once a player shoots an arrow, it travels in a line for a given distance, and may either hit an entity, bounce back from terrain, or drop on the ground once reaching the end.

If hitting an attackable entity, physical damage is inflicted. Additionally, the target will receive a small particle on top, indicating a stuck arrow. Each hit will increase an internal counter of arrows, which will all be dropped along the rest of the drops once the entity is killed and decayed.

▼ Burning terrain and units



Apart from campfires as part of NPC structures, spells and equipment can be enchanted with elemental effects, such as fire.

When a unit is hit by a fire spell, it starts to burn, periodically ticking and causing 1 true damage (ignoring resistances). If during the burn duration the unit steps into water, the fire is extinguished, smoke appears, and no more damage is dealt.

While standing on top of a campfire, 1 true damage is dealt every world tick, and the unit will remain burning for a short while after leaving the fire.

Burnings plants causes them to combust after a slightly randomized time. Trees will leave burnt stems behind, whereas grass and bushes will quickly fade away completely. Units such as hedges will not drop leaves.

Fire spells touching ice will thaw it, turning it back to water and creating a bit of smoke.

Rocks, sand and mountains are unaffected by fire.

▼ Collecting items



Items can be held by players and NPCs in various ways.

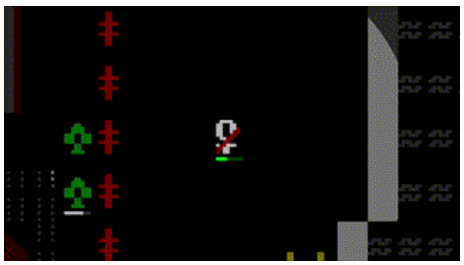
Up to 99x “countables” of each type can be held, that includes: gold, ore, stick, berry, flower and leaf. They also have different sprites if appearing as 1x, 2x or 3x and more. When collecting a stash of more than 3x countables, up to 3x at a time can be collected.

Certain item types are “stackable” in groups of 10, such as apple, banana, arrow, iron, seed, essence. The current amount of the stack is indicated by small white dots below.

Potions might have multiple charges but still count as a single item.

All other items, such as equipment, spells or tools occupy one inventory slot. Collecting an item that can be equipped while already holding one, will cause them to be replaced, dropping the previous one on the floor next to the entity. Therefore it is not possible (and also not necessary) to select and equip items in a separate screen, only one variant of an equipment slot can be taken at any given point in time.

▼ Consumption of potions and reinforcements



Any potions held in the inventory are consumed automatically, no action needed or possible by the player.

If the HP or MP is below its maximum minus a threshold of 2-3, the lowest percentage stat is chosen and a consumption is queued. The threshold allows for the player to collect dropped HP and MP from mobs, without wasting precious potions.

Different potions have different cooldowns but range from 2-3 seconds. Only one potion will be queued at a time.

Once the cooldown has passed, one charge from the potion is removed from the inventory and an animation appears above the player, receiving the target stat once the item has reached back to the player. Then, a new consumption is queued if necessary.

Health potions hold 2-3x larger amounts than mana potions.

▼ Melee damage



If an entity holds a sword and moves into an entity that can be attacked, it inflicts physical damage.

When receiving damage as the current player, the screen shakes depending on the amount of damage received.

Physical attack is calculated by the “power” stat plus the strength of the sword of the attacker.

Physical defense is calculated by the “armor” stat plus the strength of the shield worn by the defending entity.

The resulting damage is attack subtracted by defense. If however the resulting amount is below 1, an inverse fraction of damage is applied, meaning that 1 dmg will be applied every X hits.

```
const sword = attacker[EQUIPPABLE].melee;
const attack = attacker[STATS].power + sword[ITEM].amount;

const shield = defender[EQUIPPABLE].shield;
const defense = defender[STATS].armor + shield[ITEM].amount;

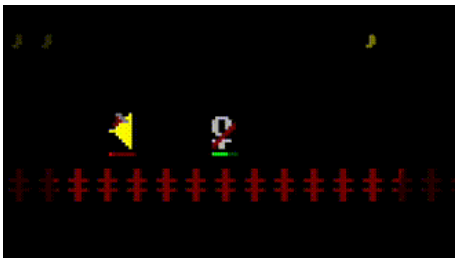
const amount = attack - defense;

if (amount > 0) return amount;

return 1 / (2 - amount);

// 2 atk - 0 def =   2 dmg
// 5 atk - 4 def =   1 dmg (1 dmg every hit)
// 4 atk - 4 def = 1/2 dmg (1 dmg every 2 hits)
// 3 atk - 4 def = 1/3 dmg (1 dmg every 3 hits)
// 2 atk - 4 def = 1/4 dmg (1 dmg every 4 hits)
```

▼ Dropping items



Entities can have an inventory and equipped items. Each item can be marked as “bound”, in which case it will not be dropped on death.

When an entity dies, a decay particle is shown (black overlay). After a short delay, all items are dropped closest to the entity’s position, gradually increasing in range.

If a cell is not walkable, no item will be dropped there. Also the direction of dropping items is randomized, so in the case of dropping a single item it may drop upwards, sidewards or downwards in any direction.

When an item is dropped, it is animated at a fixed speed towards its target location. Once it has reached its location, it may be picked up by another entity with an inventory.

▼ Entering buildings



Structures composed of multiple cells, such as a house, can be marked as a building. Once the player steps into the boundaries of it, by opening and entering through the door, the camera is fixed to the center of the building.

Once inside, all cells in the building will be revealed, and all cells outside will be hidden. Only particles will still be visible.

The cells of the building itself usually render differently when inside, this is to give an impression of seeing walls from the inside vs. from the outside. Similarly, the roof disappears.

▼ Life and death



If the player is killed, all items are dropped. A gravestone is created on its last position.

After showing a “RIP” dialog, the player can use the primary action **SPAWN** to initiate the respawn sequence. Once triggering it, a ghost with halo will appear, which will fly towards the last spawn location.

Once it arrives, the player is revived, without any items or equipment except the compass. The waypoint marker in grey is pointing towards the location of the grave, where all items can be collected.

All items can be retrieved again except for current mana and XP.

▼ Spinning focus



The focus highlight can be used in quests to indicate the next action. It spins at a relatively fast pace and does not interact with the world otherwise.

It can be green (for waypoints or collecting items), red (to kill something) or grey (to point to the gravestone).

▼ Freezing terrain and units



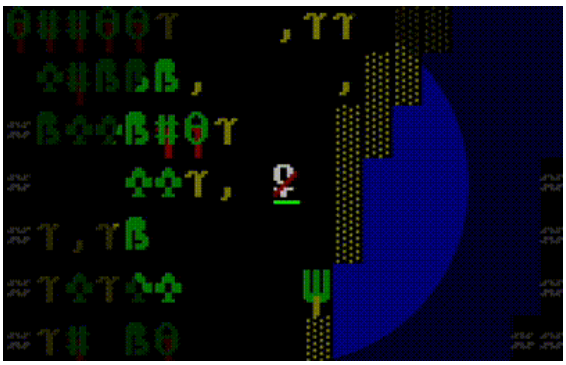
Similar to the fire element, spells can be enchanted.

Any water hit by a water spell will freeze and turn into ice. Stepping on the ice will cause the entity to slide and be unable to interact until hitting a wall or sliding out of the ice area.

Entities hit by an water spell will freeze as well for a certain duration, being stunned and unable to interact for the entire duration.

In case a frozen entity is hit by a fire spell, it will unfreeze.

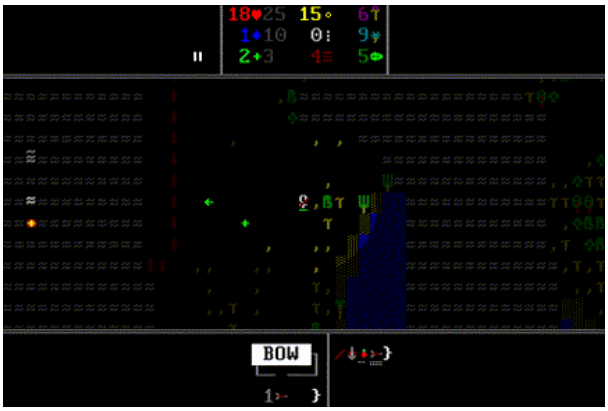
▼ Immersing in water



Ground based entities that can walk will sink into water upon stepping into it. When reaching land, the entity will appear normally again.

While walking through water, small bubbles appear in the path behind the entity.

▼ Leveling 🚧



Mobs sometimes drop XP which can be collected by the player. When reaching the maximum XP for the current level, an animation is played, the level is increased, the current XP is reset and a new, higher maximum XP is set.

After a short delay, a few stat increase items are dropped, for additional max HP and max MP. The amount of max stat items increases with each level.

▼ Magic spells affecting entities



Each time a spell is casted, certain cells will cause units to take damage, heal, burn or freeze. However, every entity can only be affected once by each time a spell is casted.

One spell can affect as many entities as long as it is active.

Entities that are friendly to each other will not have any effect for damaging, burning or freezing spells, but will trigger healings. Same way around, hostile units will take damage, burn or freeze but ignore any healing spells from enemies.

▼ Fog of war

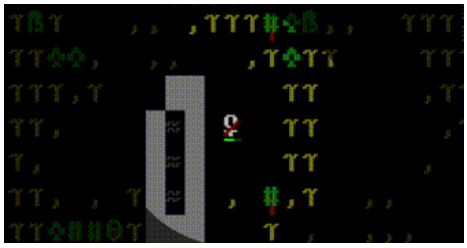


Each cell can have three fog of war states: “hidden”, “fog” and “visible”.

Initially, the whole map is covered in fog. When the player walks through the map, each cell is updated if it visible by the player. Mountains obstruct the field of vision.

Once a cell is revealed but it gets out of vision range, the terrain and certain particles on it will remain visible, like fire and smoke. Item drops, units, NPCs and mobs however will not be visible.

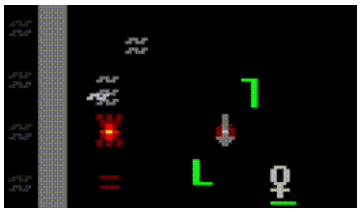
▼ Movement of entities



All entities that can move, such as the player, NPCs or mobs can perform one action each tick. If no other action like spells or attacks are performed, the entity moves in one of four linear directions by one cell.

Certain terrain and structures are not walkable, whereas other terrain and plants can be passed through, like grass, bushes, sand and water (along the shores).

▼ Compass needle

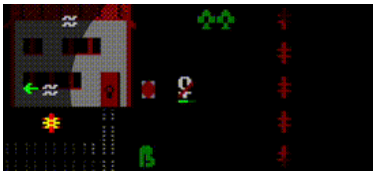


The compass always has a target, and the needle points to the direction closest to the angle towards it.

While it’s on the ground, it will point to the player, and update accordingly.

When collected and visible in the inventory, it will point to the player's spawn point.

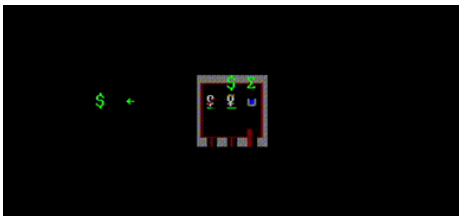
▼ Pushing boxes



Certain units, such as boxes, can be pushed around by the player. Only if the cell behind is free, they will move, meaning that two adjacent boxes can't be pushed.

Since moving into an entity usually triggers an attack, pushable entities can't be attacked with swords.

▼ NPC shops



Most NPCs will offer quests or items to trade. When standing close to the NPC, the primary action **SHOP** can be triggered, which opens a dialog box displaying all available items and their prices.

The usual movement controls turn into shop controls, up and down, and by using the primary action again an available trade can be performed.

Pressing the secondary action will close the shop.

▼ NPC crafting 🚧

(currently looks like NPC shops)

Due to the multitude of combinations between craftable and enchantable items and the ingredients, displaying a list of all options would be too large. Therefore the item and the ingredient are selectable independently, showing a resulting item, which can then be confirmed.

▼ Spikes dealing damage



Units such as cacti can have spikes, which deals damage to any entity touching or attacking it.

The damage is also dealt when not wielding a sword, and will happen before the unit takes damage. That means if both attacker and spiky unit have only 1 HP left, the attacker will die before being able to damage the unit.

Spikes deal true damage, ignoring any shields and armor stat.

▼ Dialogs and tooltips 🚧



Entities and units usually have tooltips as opposed to terrain.

NPCs or structures can also have an active dialog.

Both tooltip and dialog is shown when the player steps close to those entities. Only one dialog or tooltip is shown at a time, with the exception if an NPC shouts.

When there are multiple dialog lines, they appear after another after a delay.

Moving around an entity with a dialog or tooltip will change the position of it so that the player is never hidden by the text.

Drops of stackable items and equipment show tooltips, whereas stats don’t, such as HP, gold or berries.

Sequences

“Sequences” are used for high frequency operations, such as particle animations

▼ Primary spell: Beam



Creates a beam of magic balls that high deal damage to anyone passing through. Lasts for a short moment, creating a damaging barrier.

Consumes one mana.

Orb mobs use the same spell.

▼ Primary spell: Wave



Creates a diamond shaped wave that fades away at a given distance. Deals low damage to anyone passing through.

Consumes one mana.

▼ Secondary ability: Slash



Spins the currently worn sword around the player and deals 50% of total physical damage.

Consumes one charge.

When killing a mob with a melee attack, a charge is dropped, unless 10 or more charges are in the inventory.

The color of the slash reflects the material of the sword.

▼ Secondary ability: Bow

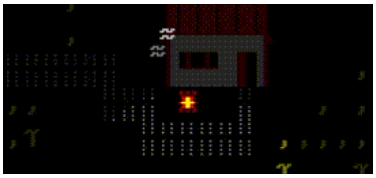


Creates a flying arrow that deals 50% of total physical damage.

Consumes one arrow.

More details can be found in “System: Ballistics of arrows”.

▼ Fire and smoke



Fire particle updates with every world tick.

Slow or fast smoke particles appear randomly and float up.

(E) Engine 🚧

Using React, TypeScript and Three.js (with react-three-fiber).

- ECS for entities and dynamics
- WFC for town generation
- 4D Perlin noise for tiling world generation
- Ticks and reference frames

TODO: Write section

(F) Fighting & Abilities

▼ Stats 🚧

Each entity, NPC, mob or unit has stats, with following base attributes:

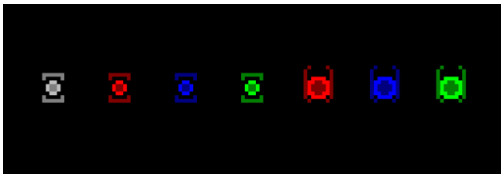
Stat name	Values	Description
Level	1 - 10	Current entity level
Max XP	3 - 50	XP needed to achieve next level
XP	0 - max XP	Current experience points
Max HP cap	40 - 60	Depending on class, the maximum HP at max level
Max HP	20 - max HP cap	The current maximum HP at current level
HP	0 - max HP	Current health points
Max MP cap	20 - 40	Depending on class, the maximum MP at max level
Max MP	5 - max MP cap	The current maximum MP at current level
MP	0 - max MP	Current mana points

For fighting, following additional stats are tracked:

Stat name	Values	Description
Power	0 - 3	Additional damage to all physical attacks
Magic	0 - 3	Additional damage to all magic attacks
Armor	0 - 3	Additional defense for all physical and magic attacks
Haste	0 - 3	Additional speed intervals

▼ Abilities

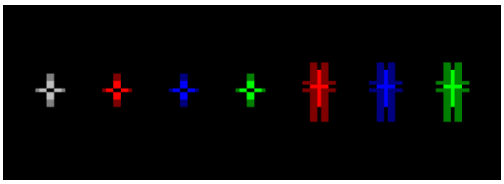
Primary spell: Wave



Creates a diamond shaped wave dealing low damage, and elemental effects.

Level 2 spell has bigger range and turns back at the middle.

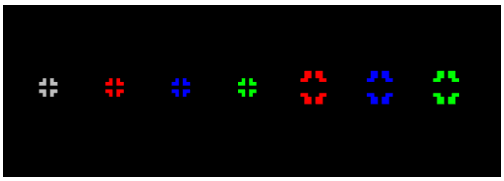
Primary spell: Beam



Creates a linear beam of magic balls dealing medium damage and elemental effects.

Level 2 spell is 3 cells wide and remains for longer.

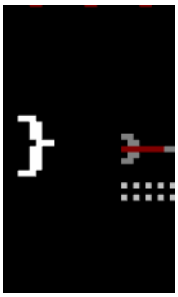
Primary spell: Trap 💡



Creates a trap on the ground dealing high damage to anyone stepping on it.

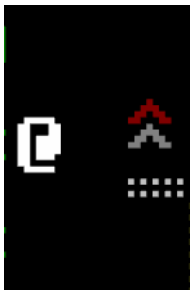
Level 2 is a plus arrangement of cells rather than one single cell.

Secondary ability: Bow



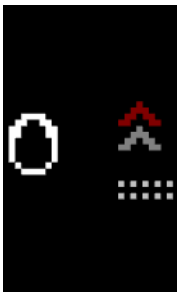
Scales damage with equipped sword

Secondary ability: Slash



Scales damage with equipped sword

Secondary ability: Block 💡



Reflects one attack

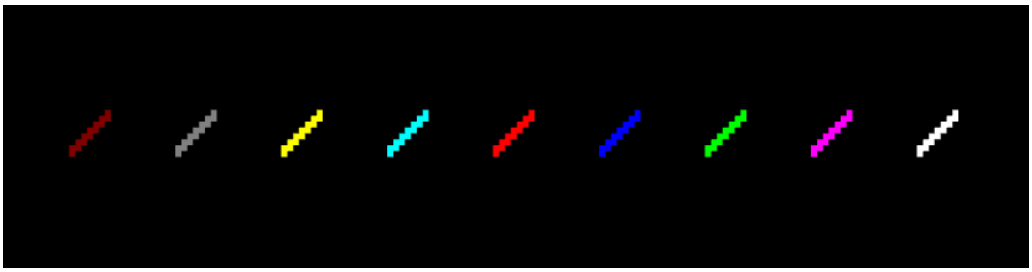
▼ Faction matrix (hostile/friendly) 🚧

Factions	Units	Wild	Neutral	Fire	Water	Earth
Units	🤝	🤝	⚔️	⚔️	⚔️	⚔️
Wild	🤝	🤝	⚔️	⚔️	⚔️	⚔️
Neutral	⚔️	⚔️	🤝	🤝	🤝	🤝
Fire	⚔️	⚔️	🤝	🤝	⚔️	⚔️
Water	⚔️	⚔️	🤝	⚔️	🤝	⚔️
Earth	⚔️	⚔️	🤝	⚔️	⚔️	🤝

(G) Gear & Crafting

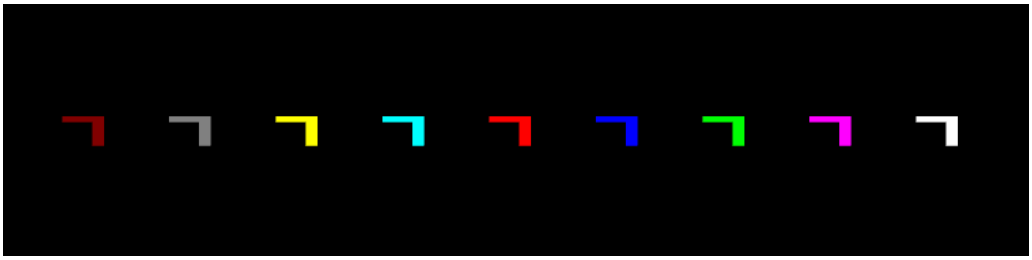
▼ Equipments

Swords



To deal physical damage. Increased amount with each tier.

Shields



Reducing any incoming physical damage, and magic effects at higher tiers.

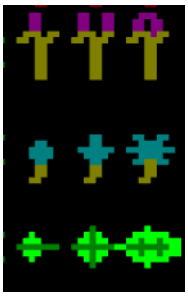
▼ Tools

Aside from fighting equipments, there are following tools:

Gear	Function
Compass	Points towards player's spawn point
Torch	Increases player's vision radius
Map 💡	Zooms out and displays a mini map

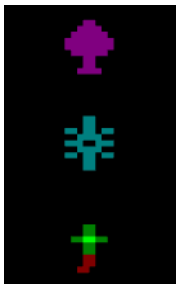
▼ Resources

Stat items



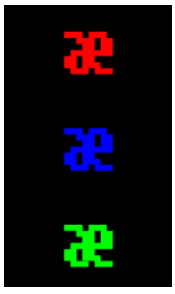
- Naturally occurring
- Berries
- Flowers

Stack stats



- Berry stack
- Flower stack
- Seed

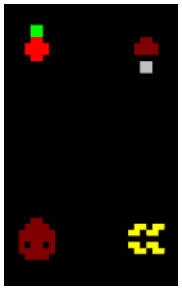
Essences



- Fire essence
- Water essence
- Earth essence

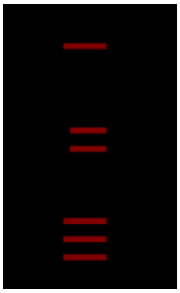
- Leaves

Fruit



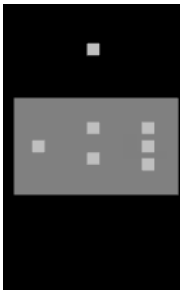
- Apple, shroom from forests
- Coconut, banana from palms

Sticks



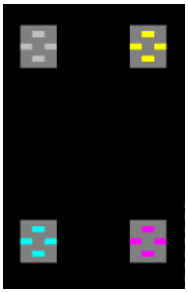
- Naturally occurring in forests

Ores



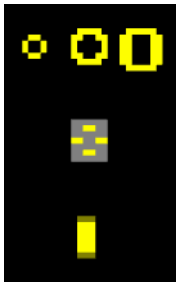
- Naturally occurring in mountains or deserts

Mines 🚧



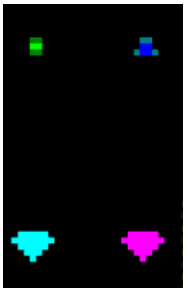
- Naturally occurring in mountains
- Iron, gold
- Diamond, ruby

Gold



- Coins, gold, ingot
- Drops from mobs
- Tradable with NPCs
- Global currency

Gems 🚧



- Gem, crystal from desert rocks
- Diamond and ruby from crafting mines

▼ Trading for resources 🚧

Item	Price options
Stick	- 3 leaf
Ore	- 2 coin
Wood	- 15 coin - 10 stick
Iron	- 20 coin - 10 ore
Gold	- 99 coin - 5 iron
Ingot	- 10 gold
Diamond	- 5 gold
Ruby	- 5 diamond

▼ Trading for elements

Item	Price options
Berry stack	- 10 coin - 10 berry
Flower stack	- 10 coin - 10 flower
Seed	- 10 coin - 10 leaf
Fire essence	- 5 berry stack
Water essence	- 5 flower stack
Earth essence	- 5 seed

▼ Trading for equipment 🚧

Item	Price options
10x Health potion	- 1 berry stack + 3 leaf
10x Mana potion	- 1 flower stack + 3 leaf
Wood sword	- 1 stick
Wood shield	- 3 wood
Wave	- 10 coin + 1 mana potion
Beam	- 10 coin + 1 mana potion
Bow	- 5 coin + 1 seed
10x Arrow	- 10 stick + 1 iron
Slash	- 5 coin + 1 seed
10x Charge	- 10 ore + 1 wood

▼ Crafting 🚧

Item	Price options
Iron sword	- 1 wood sword + 3 iron
Gold sword	- 1 iron sword + 3 gold
Diamond sword	- 1 gold sword + 3 diamond
Ruby sword	- 1 diamond sword + 3 ruby
Iron shield	- 1 wood shield + 5 iron
Gold shield	- 1 iron shield + 5 gold
Diamond shield	- 1 gold shield + 5 diamond
Ruby shield	- 1 diamond shield + 5 ruby
Fire enchantment	- 1 sword/shield + 3 fire essence
Water enchantment	- 1 sword/shield + 3 water essence
Earth enchantment	- 1 sword/shield + 3 earth essence
Fire wave	- 1 wave + 1 fire essence
Water wave	- 1 wave + 1 water essence

Item	Price options
Earth wave	- 1 wave + 1 earth essence
Fire beam	- 1 beam + 1 fire essence
Water beam	- 1 beam + 1 water essence
Earth beam	- 1 beam + 1 earth essence